BOOLEAN ALGEBRAS OF INFINITE GRAPHS AND TANGLE SPACES

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I will present the recently submitted paper I co-authored with Gustavo Boska on a Boolean algebra based approach to studying infinite graphs. In it, we define two algebras which are able to capture the connectivity structure of a given infinite graph. These two algebras, when applied the famous Stone duality functor on, give rise to a number of topological spaces that have been studied in the literature of infinite graphs for the last 60 years. This provides a new approach to defining and working with these structures. In the paper, we also pursue attempts to functorialize these constructions. I will also present how the ideas developed through the course of this paper led to the work I am developing right now on tangle theory. The abstract theory of tangles was developed by graph theorists in the last 10 years to a wide success of different applications in combinatorics. I will show how tangles are intimately connected to ultrafilters and how that can lead to a host of new and interesting questions for tangles in set theory and logic.

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References

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